

**THIRD SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION  
NOVEMBER 2020**

**Mathematics**

**ME 3C 03—MATHEMATICAL ECONOMICS**

**(Multiple Choice Questions for SDE Candidates)**

**Time : 15 Minutes**

**Total No. of Questions : 20**

**Maximum : 20 Marks**

**INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

**ME 3C 03—MATHEMATICAL ECONOMICS**  
**(Multiple Choice Questions for SDE Candidates)**

1. Curve that is also known as equal product curve is :  
(A) Indifference curve. (B) Isoquants.  
(C) Demand Curve. (D) None of the above.
2. If  $\frac{dy}{dt} = 15$ , the value of  $Y_{(t)}$  is :  
(A) 15. (B) 0.  
(C)  $15t + A$ . (D) None of the above.
3. The third stage in the law of variable proportion is called :  
(A) Increasing returns. (B) Diminishing returns.  
(C) Negative returns. (D) Proportional return.
4. In the Cobb-Douglas production function  $AK^\alpha L^\beta$ , A denotes :  
(A) Inputs. (B) Output.  
(C) Efficiency parameter. (D) None of the above.
5. Production function shows technological relationship between input and \_\_\_\_\_.  
(A) Output. (B) Factors of production.  
(C) Both of the above. (D) None of the above.
6. Marginal product equals :  
(A)  $\frac{TP}{p}$ . (B)  $\frac{AP}{p}$ .  
(C) Slope of TP curve. (D) Slope of AP curve.
7. Slope of Iso-quant is called :  
(A) MRS. (B) MRTS.  
(C) MP. (D) AP.

8. Euler's theorem is valid only for \_\_\_\_\_ function.
- (A) Non-linear. (B) Linear.  
(C) Quadratic. (D) Exponential.
9. Law of diminishing returns is also known as :
- (A) Variable proportion. (B) Returns to scale.  
(C) Isoquant. (D) Price line.
10. Higher isoquants represents higher :
- (A) Profit. (B) Output.  
(C) Cost. (D) None of the above.
11. What is the order of differential equation  $\frac{dy}{dt} = 10x + 5$  ?
- (A) First. (B) Second.  
(C) Third. (D) Fourth.
12. Given the demand curve  $P = 10 - 0.2 Q$ , total revenue curve :
- (A)  $1 - 0.2 Q$ . (B)  $10 Q^2 - 0.2 Q$ .  
(C)  $10 Q - 0.2 Q$ . (D)  $10 Q - 0.2 Q^2$ .
13. What is the degree of  $\left(\frac{dy}{dt}\right)^4 - 6t^5$  ?
- (A) First degree. (B) Second degree.  
(C) Third degree. (D) Fourth degree.
14. Demand function  $Q = f(P)$  if point elasticity  $\epsilon = -1$  for all  $P > 0$  will be :
- (A) CP. (B)  $\frac{c}{p}$ .  
(C) P. (D) None of the above.

15. What is the order of  $\left(\frac{d^2y}{dt^2}\right)^7 + \left(\frac{d^2y}{dt^3}\right)^6 = 100y$  ?
- (A) 2. (B) 3.  
(C) 5. (D) 7.
16. The degree of  $\frac{d^2y}{dt^2} + \left(\frac{dy}{dt}\right)^3 + 25 = 0$  is :
- (A) First. (B) Second.  
(C) Third. (D) Fourth.
17. Law of variable proportion explains for :
- (A) Short run. (B) Long run.  
(C) Medium. (D) None of the above.
18. In the Cobb-Douglas production function  $AK^\alpha L^\beta$ ,  $\alpha$  denotes :
- (A) Labour share. (B) Capital share.  
(C) Output. (D) Input.
19. If both factors  $X_1$  and  $X_2$  are perfect substitutes, then the value of elasticity of substitution is :
- (A) 1. (B) 0.  
(C)  $0 < \sigma < 1$ . (D) Infinity.
20. Which of the following is not a feature of Cobb-Douglas production factor ?
- (A) Linearity. (B) Homogeneity.  
(C) Constant returns. (D) Increasing returns.

**THIRD SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION  
NOVEMBER 2020**

Mathematics

ME 3C 03—MATHEMATICAL ECONOMICS

Time : Three Hours

Maximum : 80 Marks

**Part A**

*Answer all the twelve questions.  
Each question carries 1 mark.*

1. What is the order of the differential equation  $\frac{d^2y}{dx^2} + \frac{d^4y}{dx^4} - \left(\frac{dy}{dx}\right)^2$  ?  
(a) 4.                      (b) 3.                      (c) 2.                      (d) 1.
2. Write a differential with first order and 2nd degree.
3. Give an example of exact differential equation.
4. In the Cobb-Douglas production function  $AK^\alpha L^\beta$ , A denotes :  
(a) Inputs.                      (b) Output.  
(c) Efficiency parameter.                      (d) None of the above.
5. When the total product is maximum, marginal product will be :  
(a) Minimum.                      (b) Zero.  
(c) Maximum.                      (d) Negative.
6. Slope of ISo-quant is called :  
(a) MRS.                      (b) MP.  
(c) AP.                      (d) MRTS.
7. Technical relationship between input and output is called :  
(a) Elasticity.                      (b) Production function.  
(c) Input function.                      (d) None of the above.
8. Demand function  $Q = f(P)$ , the variable P denotes :  
(a) Product.                      (b) Production.  
(c) Price.                      (d) Profit.

**Turn over**

9. In CES production function, the elasticity of substitution is :  
 (a) Unity. (b) Zero.  
 (c) Negative. (d) Constant.
10. A line that connects various equilibrium points \_\_\_\_\_ of producer is :  
 (a) Isocost line. (b) Isoquants.  
 (c) Expansion path. (d) Price line.
11. If both factors  $X_1$  and  $X_2$  are perfect substitutes, then the value of elasticity of substitution is :  
 (a) 1. (b) 0.  
 (c)  $0 < \sigma < 1$ . (d) Infinity.
12. Consumer will be at equilibrium when he maximizes :  
 (a) Profit. (b) Output.  
 (c) Satisfaction. (d) Income.

(12 × 1 = 12 marks)

**Part B**

*Answer any six questions in two or three sentences.  
 Each question carries 3 marks.*

13. Find the general solution of  $(6xy + 9y^2)dy + (3y^2 + 8x)dx = 0$ .
14. Solve the difference equation  $y_t = \frac{1}{8}y_{t-1}$ .
15. Solve  $8y_{t-2} - 2y_{t-3} = 120$  and  $y_0 = 28$ .
16. Write a short note on isoquants.
17. Explain briefly 'Law of variable proportions'.
18. What are the assumptions made in the Euler's Theorem ?
19. What are the main drawbacks of Cobb - Douglas production function.
20. Write a short note on Payback method of investment appraisal.
21. Write any three similarities between Net Present Value method and Internal Rate of Return method.

(6 × 3 = 18 marks)

**Part C**

*Answer any six questions from the following.  
Each question carries 5 marks.*

22. Solve by finding the integrating factor :  $4xdy + (16y - x^2)dx = 0$ .
23. Derive the formula for the total value of an initial sum of money  $P(0)$  set out for  $t$  years at interest rate  $i$ , when  $i$  is compounded continuously.
24. Find the firms expansion path if the production function is  $q = 12\log x_1 + 30\log x_2$  and the input price of unit values of  $x_1$  and  $x_2$  respectively are  $P_1$  and  $P_2$  and  $P_1 = 2, P_2 = 5$ .
25. Given the equation for a production isoquant  $16K^{1/4}L^{3/4} = 2144$ . Find the marginal rate of technical substitution.
26. Given  $Q = 4\sqrt{KL}$ , find (a)  $MP_K$  and  $MP_L$  and (b) determine the effect on  $Q$  of a 1-unit change in  $K$  and  $L$ , when  $K = 50$  and  $L = 600$ .
27. Use the properties of homogeneity to show that a strict Cobb-Douglas production function  $q = AK^\alpha L^\beta$ , where  $\alpha + \beta = 1$ , exhibits constant returns to scale.
28. A project of Rs. 20 lakhs yielded annually a profit of Rs. 4 lakhs after depreciation at 10% and is subject to income tax at 40%. Calculate payback period of this project.
29. Write the main advantages and limitations of Net Present Value (NPV) method.
30. The expected NPV of a project has been worked out as Rs. 23,206 and its standard deviation is Rs. 9,343. If the total distribution is approximately normal and assumed continuous, then determine the probability of NPV under the following ranges : (i) Zero or less ; (ii) Greater than zero.

(6 × 5 = 30 marks)

**Part D**

*Answer any two questions from the following.  
Each question carries 10 marks.*

31. For the data given below, determine (a) the market price  $P_t$  in any time period, (b) the equilibrium price  $P_e$ , and (c) the stability of the time path :  $Q_{dt} = 180 - 0.75 P_t$ ,  $Q_{st} = -30 + 0.3P_{t-1}$ ,  $P_0 = 220$ .
32. Given a budget constraint of Rs.108 when  $P_K = 3$  and  $P_L = 4$ . Optimize the generalized Cobb-Douglas production function  $q = AK^{0.4}L^{0.5}$ .

**Turn over**

33. Determine the Internal Rate of Return (IRR) to the nearest percent for a project requiring an initial outlay of Rs. 10,000 resulting in a cash flow of Rs. 2,000 at the end of years one through five and Rs. 5,000 at the end of year six.
34. The probability distributions of NPV's of two investment projects A and B are as follows :

| Project A |             | Project B |             |
|-----------|-------------|-----------|-------------|
| NPV (Rs.) | Probability | NPV (Rs.) | Probability |
| 5,000     | 0.20        | 0         | 0.1         |
| 7,500     | 0.70        | 7500      | 0.7         |
| 10,000    | 0.10        | 15000     | 0.2         |

- (i) Compute the expected NPV of these projects ; (ii) Compute the risk attached to each points ;  
 (iii) Which project is more risky ? Why ?

(2 × 10 = 20 marks)